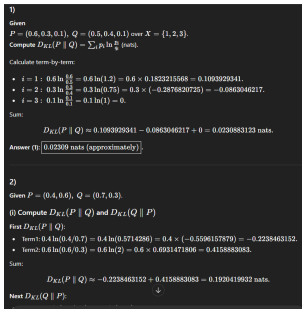
# Self-Learning: Kullback–Leibler (KL) Divergence



1)

**Given**

$P = (0.6, 0.3, 0.1)$, $Q = (0.5, 0.4, 0.1)$ over $X = \{1, 2, 3\}$.

**Compute** $D_{KL}(P \parallel Q) = \sum_i p_i \ln \frac{p_i}{q_i}$ (nats).

Calculate term-by-term:

- $i = 1$ : $0.6 \ln \frac{0.6}{0.5} = 0.6 \ln(1.2) = 0.6 \times 0.1823215568 = 0.1093929341$.
- $i = 2$ : $0.3 \ln \frac{0.3}{0.4} = 0.3 \ln(0.75) = 0.3 \times (-0.2876820725) = -0.0863046217$.
- $i = 3$ : $0.1 \ln \frac{0.1}{0.1} = 0.1 \ln(1) = 0$.

Sum:

$$D_{KL}(P \parallel Q) \approx 0.1093929341 - 0.0863046217 + 0 = 0.0230883123 \text{ nats.}$$

**Answer (1):** $\boxed{0.02309 \text{ nats (approximately)}}$.

2)

Given $P = (0.4, 0.6)$, $Q = (0.7, 0.3)$.

(i) Compute $D_{KL}(P \parallel Q)$ and $D_{KL}(Q \parallel P)$

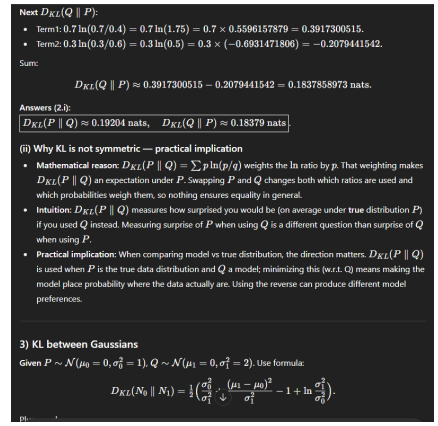
First $D_{KL}(P \parallel Q)$:

- Term1: $0.4 \ln(0.4/0.7) = 0.4 \ln(0.5714286) = 0.4 \times (-0.5596157879) = -0.2238463152$.
- Term2: $0.6 \ln(0.6/0.3) = 0.6 \ln(2) = 0.6 \times 0.6931471806 = 0.4158883083$.

Sum:

$$D_{KL}(P \parallel Q) \approx -0.2238463152 + 0.4158883083 = 0.1920419932 \text{ nats.}$$

Next $D_{KL}(Q \parallel P)$:



Next $D_{KL}(Q \parallel P)$:

- Term1: $0.7 \ln(0.7/0.4) = 0.7 \ln(1.75) = 0.7 \times 0.5596157879 = 0.3917300515$.
- Term2: $0.3 \ln(0.3/0.6) = 0.3 \ln(0.5) = 0.3 \times (-0.6931471806) = -0.2079441542$.

Sum:

$$D_{KL}(Q \parallel P) \approx 0.3917300515 - 0.2079441542 = 0.1837858973 \text{ nats.}$$

**Answers (2.i):**

$\boxed{D_{KL}(P \parallel Q) \approx 0.19204 \text{ nats}, \quad D_{KL}(Q \parallel P) \approx 0.18379 \text{ nats}}$.

**(ii) Why KL is not symmetric — practical implication**

- **Mathematical reason:** $D_{KL}(P \parallel Q) = \sum p \ln(p/q)$ weights the ln ratio by $p$. That weighting makes $D_{KL}(P \parallel Q)$ an expectation under $P$. Swapping $P$ and $Q$ changes both which ratios are used and which probabilities weigh them, so nothing ensures equality in general.
- **Intuition:** $D_{KL}(P \parallel Q)$ measures how surprised you would be (on average under **true** distribution $P$) if you used $Q$ instead. Measuring surprise of $P$ when using $Q$ is a different question than surprise of $Q$ when using $P$.
- **Practical implication:** When comparing model vs true distribution, the direction matters. $D_{KL}(P \parallel Q)$ is used when $P$ is the true data distribution and $Q$ a model; minimizing this (w.r.t. Q) means making the model place probability where the data actually are. Using the reverse can produce different model preferences.

**3) KL between Gaussians**

Given $P \sim \mathcal{N}(\mu_0 = 0, \sigma_0^2 = 1)$, $Q \sim \mathcal{N}(\mu_1 = 0, \sigma_1^2 = 2)$. Use formula:

$$D_{KL}(N_0 \parallel N_1) = \tfrac{1}{2}\left(\frac{\sigma_0^2}{\sigma_1^2} + \frac{(\mu_1 - \mu_0)^2}{\sigma_1^2} - 1 + \ln \frac{\sigma_1^2}{\sigma_0^2}\right).$$

Plug numbers:

- $\sigma_0^2/\sigma_1^2 = 1/2 = 0.5$.
- $(\mu_1 - \mu_0)^2/\sigma_1^2 = 0$.
- $\ln(\sigma_1^2/\sigma_0^2) = \ln 2 \approx 0.6931471806$.

So inside bracket: $0.5 - 1 + 0.6931471806 = 0.1931471806$. Multiply by $1/2$:

$$D_{KL} \approx 0.0965735903 \text{ nats.}$$

**Answer (3):** $\boxed{D_{KL}(P \parallel Q) \approx 0.09657 \text{ nats.}}$

(If you prefer bits: divide by $\ln 2$ to convert.)

**4) Classifier example**

**Given**

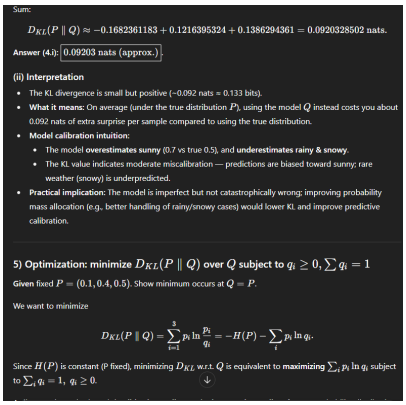
Model $Q = (0.7, 0.2, 0.1)$ for (sunny,rainy,snowy)

True $P = (0.5, 0.3, 0.2)$

**(i) Compute $D_{KL}(P \parallel Q)$**

Term-by-term:

- Sunny: $0.5 \ln(0.5/0.7) = 0.5 \ln(0.7142857) = 0.5 \times (-0.3364722366) = -0.1682361183$.
- Rainy: $0.3 \ln(0.3/0.2) = 0.3 \ln(1.5) = 0.3 \times 0.4054651081 = 0.1216395324$.
- Snowy: $0.2 \ln(0.2/0.1) = 0.2 \ln(2) = 0.2 \times 0.6931471806 = 0.1386294361$.



Sum:

$$D_{KL}(P \parallel Q) \approx -0.1682361183 + 0.1216395324 + 0.1386294361 = 0.0920328502 \text{ nats.}$$

**Answer (4.i):** $\boxed{0.09203 \text{ nats (approx.)}}$.

**(ii) Interpretation**

- The KL divergence is small but positive (~0.092 nats ≈ 0.133 bits).
- **What it means:** On average (under the true distribution $P$), using the model $Q$ instead costs you about 0.092 nats of extra surprise per sample compared to using the true distribution.
- **Model calibration intuition:**
  - The model **overestimates sunny** (0.7 vs true 0.5), and **underestimates rainy & snowy**.
  - The KL value indicates moderate miscalibration — predictions are biased toward sunny; rare weather (snowy) is underpredicted.
- **Practical implication:** The model is imperfect but not catastrophically wrong; improving probability mass allocation (e.g., better handling of rainy/snowy cases) would lower KL and improve predictive calibration.

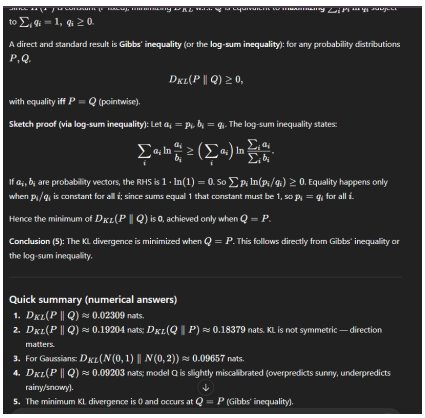
**5) Optimization: minimize $D_{KL}(P \parallel Q)$ over $Q$ subject to $q_i \geq 0, \sum q_i = 1$**

**Given** fixed $P = (0.1, 0.4, 0.5)$. Show minimum occurs at $Q = P$.

We want to minimize

$$D_{KL}(P \parallel Q) = \sum_{i=1}^{3} p_i \ln \frac{p_i}{q_i} = -H(P) - \sum_i p_i \ln q_i.$$

Since $H(P)$ is constant (P fixed), minimizing $D_{KL}$ w.r.t. $Q$ is equivalent to **maximizing** $\sum_i p_i \ln q_i$ subject to $\sum_i q_i = 1$, $q_i \geq 0$.



Since $H(P)$ is constant (P fixed), minimizing $D_{KL}$ w.r.t. $Q$ is equivalent to **maximizing** $\sum_i p_i \ln q_i$ subject to $\sum_i q_i = 1$, $q_i \geq 0$.

A direct and standard result is **Gibbs' inequality** (or the **log-sum inequality**): for any probability distributions $P, Q$.

$$D_{KL}(P \parallel Q) \geq 0,$$

with equality **iff** $P = Q$ (pointwise).

**Sketch proof (via log-sum inequality):** Let $a_i = p_i$, $b_i = q_i$. The log-sum inequality states:

$$\sum_i a_i \ln \frac{a_i}{b_i} \geq \left(\sum_i a_i\right) \ln \frac{\sum_i a_i}{\sum_i b_i}.$$

If $a_i, b_i$ are probability vectors, the RHS is $1 \cdot \ln(1) = 0$. So $\sum p_i \ln(p_i/q_i) \geq 0$. Equality happens only when $p_i/q_i$ is constant for all $i$; since sums equal 1 that constant must be 1, so $p_i = q_i$ for all $i$.

Hence the minimum of $D_{KL}(P \parallel Q)$ is **0**, achieved only when $Q = P$.

**Conclusion (5):** The KL divergence is minimized when $Q = P$. This follows directly from Gibbs' inequality or the log-sum inequality.

**Quick summary (numerical answers)**

1. $D_{KL}(P \parallel Q) \approx 0.02309$ nats.
2. $D_{KL}(P \parallel Q) \approx 0.19204$ nats; $D_{KL}(Q \parallel P) \approx 0.18379$ nats. KL is not symmetric — direction matters.
3. For Gaussians: $D_{KL}(\mathcal{N}(0,1) \parallel \mathcal{N}(0,2)) \approx 0.09657$ nats.
4. $D_{KL}(P \parallel Q) \approx 0.09203$ nats; model Q is slightly miscalibrated (overpredicts sunny, underpredicts rainy/snowy).
5. The minimum KL divergence is 0 and occurs at $Q = P$ (Gibbs' inequality).

1. Let P and Q be two discrete distributions over X = {1, 2, 3}:

P = {0.6, 0.3, 0.1}
Q = {0.5, 0.4, 0.1}

Calculate D_KL(P || Q) using natural logarithms (nats).

2. Let P = {0.4, 0.6} and Q = {0.7, 0.3}.

i. Compute both D_KL(P || Q) and D_KL(Q || P).
ii. Briefly explain why KL divergence is not symmetric and what this implies in practice.

3. Let P(x) ~ N(0, 1) and Q(x) ~ N(0, 2). Compute the KL divergence D_KL(P || Q) between these two Gaussian distributions.

Hint: D_KL(N0 || N1) = 0.5 * [ (σ0^2 / σ1^2) + ((μ1 − μ0)^2 / σ1^2) − 1 + ln(σ1^2 / σ0^2) ].

4. A classifier predicts weather categories with model distribution Q and the true distribution P:

Q: sunny (0.7), rainy (0.2), snowy (0.1)
P: sunny (0.5), rainy (0.3), snowy (0.2)

i. Compute D_KL(P || Q).
ii. Interpret what this value indicates about the model's calibration/performance.

5. You are given a fixed distribution P = {0.1, 0.4, 0.5} and must choose a distribution Q = {q1, q2, q3} to minimize KL divergence subject to q1 + q2 + q3 = 1 and q1, q2, q3 ≥ 0. Prove that the minimum KL divergence occurs when Q = P. Explain briefly why this follows from Gibbs' inequality or Jensen's inequality.